

SATURNA ISLAND EAST POINT, BC, Canada

WMO: 714730

Lat: 48.7832N

Lon: 123.0460W

Elev: 80

StdP: 14.65

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	29.0	31.7	13.8	11.1	32.8	18.2	13.8	34.6	46.8	48.4	42.3	47.9	21.0	60	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	77.4	65.3	74.6	63.9	72.1	62.5	67.2	75.2	65.1	72.5	63.3	70.4	5.1	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	88.8	72.5	61.8	82.7	69.5	60.1	77.8	67.1	31.7	75.6	30.0	72.4	28.7	70.9	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.7	29.6	24.5	DB	25.8	81.8	4.6	3.5	22.5	84.3	19.8	86.3	17.3	88.3	13.9	90.8	(4)
(5)			WB	22.6	67.7	4.5	3.3	19.4	70.1	16.7	72.0	14.1	73.8	10.8	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	43.0	43.3	46.4	50.1	55.5	59.7	63.9	63.8	59.6	52.5	46.4	42.4	(6)
(7)		DBStd	8.93	5.32	4.64	4.06	3.85	4.27	4.64	4.57	3.90	3.69	3.49	4.68	5.07	(7)
(8)		HDD50	956	220	189	122	43	6	0	0	0	0	15	123	238	(8)
(9)		HDD65	4739	681	609	577	448	294	173	75	70	166	387	559	700	(9)
(10)		CDD50	1783	4	1	10	45	177	290	430	428	288	93	14	3	(10)
(11)		CDD65	92	0	0	0	0	1	13	40	33	5	0	0	0	(11)
(12)		CDH74	307	0	0	0	0	3	53	157	90	4	0	0	0	(12)
(13)		CDH80	30	0	0	0	0	0	4	22	4	0	0	0	0	(13)
(14)	Wind	WSAvg	9.6	11.8	11.3	10.8	9.0	7.7	7.8	7.9	7.7	7.0	8.8	12.2	13.4	(14)
(15)	Precipitation	PrecAvg	33.0	4.8	2.9	3.0	2.0	1.7	1.4	0.9	1.1	1.8	3.7	4.9	4.8	(15)
(16)		PrecMax	40.8	7.6	6.1	5.9	4.6	3.0	2.7	2.0	4.2	5.4	10.2	8.5	8.1	(16)
(17)		PrecMin	23.1	2.4	0.5	0.9	0.1	0.0	0.1	0.0	0.1	0.2	0.4	1.8	1.8	(17)
(18)		PrecStd	4.6	1.4	1.2	1.2	1.0	0.8	0.7	0.5	0.9	1.3	2.0	1.9	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.9	53.7	58.1	65.0	72.3	79.5	83.1	79.9	73.9	62.9	58.5	55.3	(19)
(20)			MCWB	52.2	50.3	51.1	57.2	59.8	64.9	69.7	67.3	62.8	57.5	54.5	52.3	(20)
(21)		2%	DB	52.9	51.3	55.0	60.7	68.1	74.7	78.0	76.4	70.2	60.5	55.3	52.4	(21)
(22)			MCWB	50.5	48.2	49.9	53.5	57.7	62.9	65.8	65.6	61.1	56.2	52.8	49.7	(22)
(23)		5%	DB	51.1	49.8	53.1	58.0	65.0	70.9	75.0	73.9	67.8	58.6	53.7	50.2	(23)
(24)			MCWB	49.0	47.0	49.1	51.9	56.7	61.0	64.3	64.0	59.9	54.9	51.2	47.5	(24)
(25)		10%	DB	49.4	48.5	51.5	55.7	62.4	67.4	72.1	71.4	65.6	57.0	52.1	48.5	(25)
(26)			MCWB	47.3	46.1	48.0	50.5	55.2	59.2	62.7	62.6	58.9	53.7	49.9	46.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	51.0	53.4	58.5	62.6	67.1	70.6	69.8	64.8	59.9	55.4	53.3	(27)
(28)			MADB	54.3	52.6	55.7	63.6	68.8	76.2	80.3	78.4	71.4	61.2	56.8	54.9	(28)
(29)		2%	WB	51.2	49.1	51.4	54.7	59.6	64.2	67.9	67.0	62.8	57.5	53.2	50.0	(29)
(30)			MADB	52.2	50.8	53.5	59.2	65.4	72.7	76.2	74.4	68.2	59.3	54.9	51.9	(30)
(31)		5%	WB	49.2	47.7	50.0	52.7	57.7	61.7	65.4	64.8	61.2	55.8	51.7	48.0	(31)
(32)			MADB	50.7	49.2	52.1	56.6	63.4	69.1	72.9	72.0	66.2	57.7	53.2	49.7	(32)
(33)		10%	WB	47.6	46.5	48.6	51.1	55.9	59.8	63.3	63.0	59.8	54.2	50.3	46.5	(33)
(34)			MADB	49.2	48.1	51.0	54.9	61.3	66.5	70.6	70.1	64.5	56.4	51.9	48.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.4	6.0	7.7	9.3	11.5	12.8	14.4	13.6	10.0	6.6	5.8	5.1	(35)
(36)			MCDBR	6.7	6.8	9.3	11.6	14.8	17.3	17.6	16.7	12.9	8.1	7.1	6.5	(36)
(37)			MCWBR	5.1	5.0	6.2	7.1	8.2	8.9	8.9	9.0	7.0	5.7	5.5	5.1	(37)
(38)		5% WB	MCDBR	6.4	6.3	8.6	10.9	13.8	16.1	17.1	16.5	12.1	7.7	6.7	6.3	(38)
(39)			MCWBR	5.2	5.1	6.3	7.3	8.4	9.0	9.7	9.5	7.2	5.8	5.5	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.319	0.314	0.331	0.359	0.380	0.356	0.351	0.358	0.342	0.332	0.320	0.311	(40)	
(41)		taud	2.517	2.558	2.526	2.438	2.393	2.476	2.520	2.504	2.548	2.559	2.529	2.499	(41)	
(42)		Ebn at Noon	239	270	282	283	279	287	287	280	275	259	236	225	(42)	
(43)		Edh at Noon	19	23	28	33	36	33	32	31	27	23	19	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	285	562	866	1327	1712	1859	2047	1730	1218	669	339	228	(44)	
(45)		RadStd	41	79	108	131	140	193	157	121	103	81	56	32	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page